

SAMEDAM, Switzerland

WMO: 067920

Lat: 46.5261N

Lon: 9.8794E

Elev: 1710

StdP: 82.40

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.7	-20.6	-26.2	0.4	-20.6	-24.0	0.5	-19.9	9.0	-0.1	7.7	-2.1	1.0	200	0.723

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	23.1	13.6	21.6	12.9	20.1	12.4	14.7	20.9	13.8	19.8	13.1	18.6	5.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	10.9	16.9	11.4	10.3	15.9	10.6	9.8	15.1	47.2	21.0	44.5	19.9	42.2	18.7	18.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.4	6.5	-27.1	25.2	2.9	1.5	-29.2	26.3	-31.0	27.2	-32.6	28.1	-34.8	29.2		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-27.3	16.2	2.9	0.8	-29.4	16.8	-31.1	17.3	-32.7	17.8	-34.8	18.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-8.3	-6.9	-2.4	2.2	6.9	10.6	12.4	12.1	8.2	4.5	-1.6	-6.8
	DBStd	8.26	4.88	4.70	4.26	3.34	2.83	3.12	2.67	2.60	2.94	3.46	4.76	5.07
	HDD10.0	2921	567	474	383	234	101	31	10	11	68	172	349	521
	HDD18.3	5733	825	708	641	484	353	232	184	194	304	429	599	779
	CDD10.0	229	0	0	0	0	6	49	84	76	14	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	34	0	0	0	0	0	8	12	14	0	0	0	0
	CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0
Wind	WSAvg	2.2	1.6	1.9	2.2	2.5	2.7	2.6	2.7	2.6	2.4	2.2	1.7	1.6
Precipitation	PrecAvg	978	52	46	48	59	88	117	119	124	91	88	94	54
	PrecMax	1333	120	150	124	136	197	223	225	194	214	230	366	151
	PrecMin	738	6	13	6	5	30	52	67	20	22	1	7	0
	PrecStd	169	33	35	29	31	39	49	35	38	52	50	83	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	7.4	10.7	16.0	20.3	24.4	24.7	25.1	20.8	17.5	12.6	6.3
		MCWB	1.4	1.8	3.7	6.9	11.1	13.7	14.4	13.9	11.7	9.0	5.3	1.1
	2%	DB	2.8	4.5	8.6	13.5	17.8	22.3	23.1	22.7	18.8	15.1	8.9	3.6
		MCWB	0.1	0.1	2.5	6.0	9.7	13.1	13.9	13.6	10.9	7.8	3.9	0.9
	5%	DB	1.1	2.5	6.7	11.1	15.9	20.5	21.5	20.9	17.0	13.1	6.7	2.1
		MCWB	-0.8	-0.8	1.6	4.7	8.7	12.4	12.9	13.0	10.5	7.2	3.3	0.3
	10%	DB	-0.3	0.8	4.7	8.9	14.0	18.5	19.9	19.1	15.0	11.2	4.8	0.9
		MCWB	-1.8	-1.6	0.8	3.7	7.7	11.6	12.4	12.5	9.6	6.7	2.5	-0.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	2.2	4.4	7.9	11.9	15.4	16.0	15.7	13.3	10.3	6.7	2.7
		MCDB	4.8	6.0	9.6	13.8	18.5	22.1	22.1	22.1	18.5	14.8	9.9	4.3
	2%	WB	0.6	0.8	3.1	6.6	10.4	14.0	14.8	14.5	12.1	9.1	5.0	1.5
		MCDB	2.2	3.7	6.9	12.2	16.6	20.4	21.1	20.4	16.9	12.9	8.0	2.8
	5%	WB	-0.5	-0.3	2.1	5.3	9.3	13.0	13.8	13.7	11.2	8.2	3.7	0.6
		MCDB	0.9	1.9	5.6	10.1	14.6	19.1	19.7	19.3	15.4	11.4	5.9	1.9
	10%	WB	-1.8	-1.4	1.2	4.1	8.4	12.0	13.0	12.9	10.2	7.3	2.7	-0.5
		MCDB	-0.3	0.8	4.3	8.2	12.8	17.6	18.6	17.9	14.0	10.3	4.5	0.9
Mean Daily Temperature Range	5% DB	MDBR	12.6	14.0	12.2	10.9	11.4	12.6	13.1	12.4	12.1	12.1	10.4	11.1
		MCDBR	12.1	15.7	14.7	15.2	15.7	16.7	16.5	16.4	15.9	16.1	13.3	11.1
		MCWBR	9.4	11.6	9.2	8.5	8.6	8.6	8.2	8.3	8.8	9.4	8.6	8.2
	5% WB	MCDBR	10.1	13.6	12.3	13.6	14.1	15.1	14.7	14.4	13.5	12.3	10.7	9.2
		MCWBR	8.1	10.0	7.9	7.8	7.8	8.0	7.6	7.7	7.8	7.6	7.2	7.1
Clear-Sky Solar Irradiance	taub	0.229	0.252	0.294	0.342	0.356	0.365	0.359	0.352	0.329	0.300	0.260	0.228	
	taud	2.337	2.272	2.223	2.212	2.302	2.322	2.338	2.378	2.427	2.483	2.459	2.370	
	Ebn at Noon	895	933	932	912	909	899	899	893	887	867	849	856	
	Edh at Noon	71	95	118	132	126	124	120	111	96	78	64	62	
All-Sky Solar Radiation	RadAvg	1.62	2.58	3.81	4.49	4.83	5.62	5.64	4.90	3.81	2.54	1.57	1.34	
	RadStd	0.19	0.30	0.52	0.69	0.50	0.47	0.46	0.44	0.38	0.31	0.24	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.57	+0.58	+0.54	N/A	-144	+50	N/A
(47)	Regional (107 neighbors)	+0.33	N/A	N/A	+0.66	+0.41	+0.35	N/A	N/A	+60	+15

Nomenclature: See separate page